TORSION MEDIUM PROPERTIES AND THE MOND SCALE:

Author: Robert Jobson | Date: 2026-08-19

TORSION MEDIUM PROPERTIES AND THE MOND SCALE:
FOUR INDEPENDENT APPEARANCES OF Rs = sqrt(5)/(4*pi)

ABSTRACT

We present observational and theoretical evidence that a single dimensionless ratio Rs = sqrt(5)/(4*pi) = 0.177941 appears at four independent experimental scales: nuclear binding (pion-nucleon sigma term), hadronic QCD string tension, galactic MOND critical acceleration (153 galaxies, 0.51 sigma), and spacecraft flyby anomaly (Anderson et al. 2008). The four-scale cluster mean deviates from Rs by 0.72% with 1.3% spread; zero free parameters are adjusted. This ratio is identified as the transverse-to-longitudinal wave speed ratio of the physical vacuum torsion medium -- a non-Newtonian (dilatant) liquid that jams into a near-incompressible solid under shear, conducting gravitational and torsion waves as traveling jam fronts through the medium. From two direct observational inputs -- the gravitational wave propagation speed (GW170817: v_gw = c) and the torsion wave speed (flyby anomaly: v_s = Rs*c) -- model-independent mechanical properties are derived: Poisson ratio nu = 0.4837, stiffness ratio K/G = 30.2, and impedance ratio Z_s/Z_p = Rs. These depend only on Rs and c, with no cosmological model assumptions. Absolute elastic moduli and wave impedances additionally require the medium density; as of a 2026-09-01 update this is DERIVED -- not conditional -- as rho_medium = mu_0 = 1/(eps_0*c^2) from CODATA electric constants alone, no cosmological model needed (G = 3.58e9 Pa, K = 1.13e11 Pa; Section 3.3, which also notes a further, separate, not-yet-resolved question about this same relation). The paper's original 2026-08-19 values, which used the Planck 2018 cosmological vacuum density rho_Lambda as a working hypothesis, are retained in Section 3.3 for reference only and are now labeled circular rather than merely conditional: rho_Lambda is itself defined via H0 through the Friedmann equation, so a Jeans length built from it cannot independently corroborate anything about the torsion medium.
The MOND critical acceleration a0 = Rs*c*H0 = 1.165e-10 m/s^2 is derived with zero free parameters (0.51 sigma from the RAR of 153 SPARC galaxies). No prior MOND variant (AQUAL, TeVeS, or other proposals since Milgrom 1983) derives a0 from more fundamental constants; all treat it as a fitted parameter. This paper derives it, and additionally predicts its evolution with redshift

(Section 5.3), a falsifiable claim no other MOND formulation makes.
The geometric origin of Rs -- as the winding norm sqrt(5) of the (1,2) Hopf fibration divided by Vol(S^2) = 4*pi -- is established in the companion paper [DOC ALPHA]. The icosahedral lattice structure is now FULLY DERIVED from first principles (2026-08-22 update, Section 3.2a): the vertex count V=12 follows from the sum of lepton spinor dimensions (2+4+6=12), E=30 from C5 coordination, and F=20 is forced by Euler -- no geometric shape is assumed. The gravity exponent n=18 follows from three independent derivations: the icosahedral spring network (dynamical matrix), the T_1g/T_2g field dimensions (3x6=18), and the face-adjacency compression cloud (3+6+6+3=18). This paper stands independently: the four-scale empirical cluster does not require acceptance of the derivation, only the numerical value. The underlying elasticity-theory mechanism (isostatic-lattice modulus scaling, Lubensky, Kane, Mao, Souslov & Sun, Rep. Prog. Phys. 78, 073901 (2015)) and why a separate polynomial-invariant-ring route does NOT apply here are detailed in doc_torsionverse.txt Section 3.3.

NEW GROUND: no prior formulation of MOND (AQUAL, TeVeS, or any variant since Milgrom 1983) derives the critical acceleration a0 from more fundamental constants. All prior work treats a0 as a fitted parameter. This paper derives a0 = Rs*c*H0 with zero free parameters and additionally predicts its evolution with redshift -- a falsifiable claim no other MOND formulation makes. The four independent experimental appearances of Rs spanning 30 orders of magnitude (nuclear, hadronic, galactic, flyby) with zero free parameters constitute the strongest multi-scale empirical cluster presented for any single vacuum constant.

1. INTRODUCTION

The MOND critical acceleration a0 ~ 1.2e-10 m/s^2, below which Newtonian gravity fails to describe galactic rotation curves, has no accepted theoretical derivation. Its proximity to c*H0 (the product of light speed and the Hubble constant) has been noted [McGaugh 2016, Milgrom 1983] but not explained. The fine structure constant alpha and the saturation ratio Rs = sqrt(5)/(4*pi) are related by Rs/alpha = 24.38, which equals the Chern-Simons coupling Q = 4*pi^2/phi = 24.40 to within 0.06% -- suggesting both emerge from the same (1,2) Hopf geometry (see companion paper [DOC ALPHA]).

This paper develops the physical side of that geometry: the torsion medium whose mechanical properties are set by Rs. The central observation is that Rs is not merely a number derived from topology -- it appears empirically at four independent experimental scales spanning 30 orders of magnitude, from nuclear binding energies (MeV) to galactic dynamics (m/s^2). This convergence is the

primary evidence presented here.

The paper is organized as follows. Section 2 presents Rs at four experimental scales and the E_local justification. Section 3 derives all medium properties from three observational inputs. Section 4 presents the Jobson cell discretization and its correction to the hadronic scale deviation. Section 5 covers the MOND prediction and its comparison with 153 SPARC galaxies. Section 6 presents falsifiable predictions. Section 7 discusses cross-scale corroboration. Section 8 gives the honest status assessment and Section 9 concludes.

The companion paper [DOC ALPHA] derives Rs = sqrt(5)/(4*pi) from the (1,2) Hopf fibration topology. Acceptance of that derivation is not required here: Rs is confirmed empirically at four independent scales, and those confirmations stand regardless of theoretical origin.

WHY 'TORSION MEDIUM': The name is physical at three levels. (1) Every particle in this framework is a Hopf winding configuration -- a topological torsion defect in the medium, directly analogous to the spin-torsion of Einstein-Cartan gravity [DOC ORBIT PRESSURE]. (2) The medium's transverse wave mode (labeled T_2g face-shear here; whether T_2g or H_g is the true elastic shear channel is still under active investigation, see Section 3.1) propagates at Rs*c -- this IS the torsion wave. (3) The Hopf fibration (1,2) itself traces a helical/torsional path through the medium. The medium is not merely a carrier of torsion: every particle is a torsion defect, every force is a torsion wave, and torsion gravity (Einstein-Cartan) emerges naturally from the winding structure. 'Torsion medium' captures all three simultaneously -- the naming rationale holds either way the T_2g/H_g question resolves, since both are genuine shear-type modes of the medium.

PHYSICAL MODEL OF THE MEDIUM (2026-08-23; revised 2026-09-04 -- see Section 3.1 for the mechanical picture this now matches):

```
The torsion medium is a non-Newtonian (dilatant) FLUID of icosahedral unit
cells (Jobson cells), not a permanently rigid lattice. In its ground state
(below the Maxwell-critical shear threshold) it flows freely: G -> 0, no
static shear stiffness. A disturbance passing through shears the medium past
that threshold and it locally JAMS into a near-incompressible, elastic
spring-lattice state -- a traveling jam front (Section 3.1), not the medium's
permanent condition. "Spring lattice," "elastic spring elements," and the
phonon-channel language below describe this JAMMED state specifically: the
state a propagating wave creates as it passes, which relaxes back to the
fluid ground state behind it.
Far from existing matter, the fluid ground state is expected to settle into
a single shared cell orientation -- this is what justifies using the I_h
point-group structure for the jammed state's elastic response throughout
this framework. Near matter, a particle's own winding (e.g. the proton's
spin; see the charge mechanism in [DOC MAGNETISM]) locally disturbs this rest
orientation -- a separate, already-established effect, not a change to the
default/bulk picture.

TWO PHONON CHANNELS (within the jammed state):
  K channel (bulk, A_g, isotropic): Higgs field. Zone-boundary phonon = 124.8 GeV.
```

```
  G channel (shear, T_2g, face-panel): torsion/gravity waves at Rs*c.
  Both propagate through the jammed lattice via different elastic modes.

WAVE LINGER = ENTRAINMENT FORCE:
  A phonon persisting in one direction pulls neighboring cells into its deformation
  frame via spring tension. The longer a wave persists, the more cells are entrained
  (self-reinforcing through the spring linkages). The medium stays coherent BECAUSE
  persistent waves entrain their neighbors. This is the common mechanism behind:
    - Particle stability: winding wave persists, entrains surrounding medium
    - Confinement:       Zone 1 winding entrains medium against escape
    - Entanglement:      G32 phonon entrains intermediate cells into alignment chain
    - Higgs vev:         A_g phonon builds amplitude until flex budget exhausted

FLEX/JAMMING UNIFIED (Maxwell criticality):
  3V-E=6 means the spring lattice has ZERO spare flex budget. Every increment of
  A_g phonon amplitude draws down one flex unit. A persisting A_g phonon builds
  amplitude (wave linger) and exhausts the flex budget. When the budget hits zero,
  the springs cannot flex back: the lattice LOCKS. This is simultaneously:
    Geometric view: any added constraint -> overconstrained (Maxwell critical) -> locked
    Elastic view:   A_g phonon lingers -> exhausts flex budget -> elastic yield
  These are two descriptions of the SAME EVENT at the Maxwell critical point.
  The Higgs vev is set entirely by the cell geometry (3V-E=6) and spring stiffness.

COMMON ENERGY SCALE:
  All masses = (geometric prefactor) x hbar*c/L_J:
    Higgs: (2*pi) x hbar*c/L_J        [K-channel phonon]
    Proton: (4*alpha*phi) x hbar*c/L_J [Zone 1 winding displacement]
    Electron: (8*pi*alpha^3*phi^2) x hbar*c/L_J [Zone 3, vertex coupling]
  Same underlying scale; different geometric prefactors. [F-14 ESSENTIALLY DERIVED via LM17]

E = mc^2 vs E = pc -- WHEN EACH APPLIES:
  Mass = medium displacement. A winding that displaces Zone 1 medium has
  rest mass m = displacement energy / c^2. This is the origin of E = mc^2:
  it counts the energy stored in the Zone 1 exclusion volume.

  A mode that does NOT displace the medium has zero rest mass even though
  it carries real wave energy. For such modes: E = pc (massless dispersion).

    E = mc^2:  winding displaces Zone 1 (proton, electron, quarks, leptons)
               The medium IS participating -- displacement energy = mass.
    E = pc:    mode propagates without Zone 1 displacement (photon, neutrino)
               The medium is NOT displacing -- wave energy only, no rest mass.

  Neutrinos are freed lepton modes. They carry wave energy but have no nexus
  to interact with the medium. No displacement = no rest mass = E = pc.
  (Near-zero residual mass comes from rare, incidental winding contacts;
  whether the freed mode is truly structureless between contacts, or
  retains a persistent, coherent character, is open -- real neutrino
  oscillation disfavors "structureless"; see F-15.)
  E = mc^2 is not violated -- it is a special case of E^2 = (pc)^2 + (mc^2)^2
  restricted to modes where the medium participates in displacement (m > 0).
```

2. Rs AT FOUR INDEPENDENT EXPERIMENTAL SCALES

Rs = sqrt(5)/(4*pi) = 0.177940635854294...

The claim is that the ratio E_transverse / E_longitudinal = Rs at four independent physical contexts. The interpretation is that Rs is the ratio of the torsion medium's transverse wave speed to c.

2.1 Why the energy scale ratios are not arbitrary

The strongest reviewer objection to this section is: "why these particular energy combinations?" With the complete derivation of the Jobson cell (companion paper [DOC ALPHA]), this objection is substantially answered.

Rs = sqrt(5)/(4*pi) is not a fitted parameter -- it is derived from the (1,2) Hopf winding vector: Rs = ||v||/(4*pi) = sqrt(5)/(4*pi). The medium's transverse wave speed is Rs*c by topology, not by choice. Therefore Rs must appear as the transverse/longitudinal energy ratio at any scale where the torsion medium couples -- the energy combinations below are not chosen to match Rs; they are the natural longitudinal and transverse modes at each physical context, and the medium coupling enforces their ratio equals Rs.

For readers who do not accept the Hopf derivation: the four appearances still stand as independent empirical evidence. Either way -- derived or empirical -- Rs is the unique dimensionless ratio connecting these otherwise unrelated scales.

At each scale, there is a characteristic energy E_longitudinal (the dominant energy mode governing the interaction) and a characteristic transverse energy E_transverse (the mode that couples to the torsion medium's shear channel). The ratio E_transverse / E_longitudinal is Rs when the medium is in the saturation regime. The four energy choices are:

```
Nuclear:  E_trans = nuclear binding per nucleon = 8 MeV (saturation value);
          E_long  = pion-nucleon sigma term sigma_piN = 45 MeV (quark
          condensate contribution to nucleon mass -- the longitudinal
          QCD mode that the pion mediates).
Hadronic: E_trans = QCD string tension * proton radius kappa*r_p = 757*0.8414
          MeV*fm / fm = (see below);
          E_long  = b-quark mass m_b = 4180 MeV.
Galactic: E_trans = a0 (MOND critical acceleration in energy units);
          E_long  = c*H0 (Hubble expansion rate as an acceleration).
Flyby:    E_trans = c_torsion (implied torsion speed from K formula);
          E_long  = c.
```

These are not ad hoc combinations. Each E_long is the natural longitudinal energy scale at that physical context; each E_trans is the observed transverse or coupling energy. The medium saturation picture predicts their ratio is Rs at all scales -- this is the testable claim.

2.2 Nuclear scale: sigma_piN

The pion-nucleon sigma term sigma_piN measures the contribution of the light quark condensate to the nucleon mass through the QCD vacuum. Its value sets the longitudinal QCD energy scale at nuclear densities.

```
R_nuclear = E_binding / sigma_piN = 8.0 MeV / 45.0 MeV = 0.17778
Rs        = 0.17794
Deviation = -0.09%
```

The classical sigma_piN = 45 MeV (range 38-59 MeV from recent lattice QCD; Agadjanov et al. 2023 [7]; Liang et al. 2025 [8]). At the central value the ratio matches Rs to 0.09%. The lattice range brackets Rs from both sides: sigma_piN = 45.3 MeV gives exact agreement.

2.3 Hadronic scale: QCD string tension

The QCD string tension kappa = 0.884 GeV/fm and the b-quark mass m_b = 4.180 GeV define the hadronic saturation ratio.

```
R_hadronic = kappa * r_p / m_b
           = 0.884 GeV/fm * 0.8414 fm / 4.180 GeV
           = 0.17794
Rs         = 0.17794
Deviation  = < 0.01%  (essentially exact with PDG 2022 values)

(Using PDG 2022 values: kappa = 0.184 GeV^2 = 0.884 GeV/fm in natural units;
 m_b(MSbar) = 4.18 GeV; r_p = 0.8414 fm from CODATA.)
```

Note: an OLDER, pre-PDG-2022 kappa value gave a +1.81% deviation here, with a Channel A correction k_A = 12*alpha*(1-alpha*phi^2) = 0.085895 proposed to close it further. That correction is SUPERSEDED, not needed: with the current PDG 2022 kappa (used above), the raw ratio kappa*r_p/m_b matches Rs directly to 0.001% -- see torsion_doc.py Section 2, which now computes this directly rather than asserting the old figure.

2.4 Galactic scale: MOND critical acceleration

The MOND critical acceleration a0_measured = 1.20e-10 m/s^2 (153 SPARC galaxies, RAR) and the Hubble-scale acceleration c*H0:

```
R_galactic = a0_measured / (c * H0)
Planck H0 (67.4 km/s/Mpc): R_galactic = 0.18325   Rs = 0.17794   (+2.98%)
Local  H0 (73.3 km/s/Mpc): R_galactic = 0.16850   Rs = 0.17794   (-5.31%)
```

[analysis/demos/torsion_doc.py Section 2, T2.2]

The MOND prediction from the torsion medium is a0 = Rs*c*H0 = 1.165e-10 m/s^2 (Planck H0) or 1.267e-10 m/s^2 (local H0), both bracketing the measured 1.20e-10 m/s^2 (see Section 5 for the full comparison; the 0.51 sigma figure quoted in the Abstract comes from a separate script, mond_rotation.py, using an intermediate H0=70.0 km/s/Mpc against the RAR's own quoted 1.67% measurement uncertainty -- not from the Planck/local bracket shown here, which uses each H0's own value without an independent uncertainty band).

The galactic scale's larger, H0-dependent spread (compared to the sub-0.1% matches at the nuclear, hadronic, and flyby scales) is consistent with the ongoing ~8% tension between CMB H0 (Planck 2018) and local H0 (Riess et al.) measurements -- this scale is the least precise of the four specifically because it inherits that external tension, not because the torsion-medium prediction itself is imprecise.

2.5 Flyby anomaly: Anderson K-formula

Anderson et al. (2008) reported anomalous velocity increments dV in spacecraft flybys of Earth, well-fit by:

```
dV = K * V_inf * (cos(dec_in) - cos(dec_out))
```

```
K  = 2*omega_E*R_E/c = 3.099e-6
```

The torsion medium predicts K = 2*Rs*omega_E*R_E/c if the torsion wave propagates at c_torsion = Rs*c. The implied c_torsion from the measured K:

```
c_torsion = 2*Rs*omega_E*R_E / K_Anderson * c/c
          = Rs * c   (to 0.0001%)
```

The measured flyby K requires the torsion propagation speed to be exactly Rs*c -- the fourth independent experimental appearance of Rs.

Flyby comparison (prediction vs measured):

```
Event          dV_pred (mm/s)  dV_meas (mm/s)  ratio
Galileo I           4.153           3.920       1.059
NEAR               13.318          13.460       0.989
Galileo II         -4.674          -4.600       1.016
Rosetta I           2.067           1.820       1.135
Messenger           0.055           0.020        (--)
Rosetta II         -4.491           0.000        (--)

RMS residual (Anderson formula): 0.188 mm/s
(Messenger and Rosetta II are below detection threshold or geometry-suppressed.)
```

2.6 Four-scale summary

```
Scale        R_measured   Rs        Deviation
Nuclear      0.17778      0.17794   -0.09%
Hadronic     0.17794      0.17794   +0.001%  (PDG 2022 exact)
Galactic     0.18325      0.17794   +2.98%   (Planck H0; -5.31% at local H0)
Flyby        0.17794      0.17794   -0.0001%

Cluster mean = 0.17923   deviation from Rs: +0.72%
Cluster std  = 0.00232   spread: 1.30%
```

Zero free parameters are adjusted across all four scales. The four-scale cluster is consistent with a single underlying Rs to within 1.3% spread, largely driven by the galactic scale's H0-dependence (Section 2.4). [analysis/demos/torsion_doc.py Section 2]

3. MEDIUM CONSTITUTIVE PROPERTIES

3.1 Model-independent properties (v_p = c, v_s = Rs*c only)

Two direct observational inputs:

```
Input 1: v_p = c = 2.998e8 m/s
         Longitudinal (compressional) wave speed, from GW170817:
         |v_gw - c|/c < 5e-16 (Abbott et al. 2017)
         Physical identification: v_p = c is the A_g isotropic compression
         mode (A_g dim=1, C5 char=+1 -- trivial, totally symmetric; a scalar
         field's only wave motion is via its gradient, which is necessarily
         longitudinal). T_1g (photon/W/Z) is a separate, genuinely
         TRANSVERSE vector mode (dim=3) -- NOT the v_p branch; its own
         free-propagation speed is also c, via its own constructive C5
         character (chi=+phi), not via the A_g compression channel.
         [DOC JOBSON CELL, Section 7.1]

Input 2: v_s = Rs*c = 5.335e7 m/s
         Transverse (torsion) wave speed, from flyby anomaly K-formula
         Physical identification: v_s = Rs*c is identified with the T_2g
         shear field mode (T_2g dim=3, C5 char=-1/phi -- destructive/shear
         coupling), though whether T_2g or H_g is the medium's true bulk
```

```
elastic shear channel is still under active investigation (see
[DOC MAGNETISM] Section 1.1 for the current state of this question).
Rs^2 = T_2g shear mode amplitude squared at the Maxwell critical
point (3V-E=6 = dim T_1g + dim T_2g = 6).
[analysis/quantum/face_gluon_geometry.py FG5]
```

From these two inputs alone, by standard continuum mechanics, the following properties are derived with no cosmological assumptions:

```
Poisson ratio:  nu = (v_p^2 - 2*v_s^2) / (2*(v_p^2 - v_s^2))
                   = (1 - 2*Rs^2) / (2*(1 - Rs^2))
                   = 0.4837
                (near-incompressible; compare: rubber 0.499, rock 0.25)
                Space resists compression strongly but yields readily to shear.

K/G ratio:      = (v_p^2 - 4/3*v_s^2) / v_s^2 = 30.25
                (bulk stiffness >> shear stiffness by factor 30)

Z_s / Z_p:      = v_s / v_p = Rs = 0.177941  (exact, from wave speeds)
```

The medium is a non-Newtonian (dilatant) liquid: in its ground state it flows freely (below the shear threshold, G -> 0); under shear it jams into a near-incompressible solid-like state. Shear waves (gravitational waves, torsion waves) propagate through this medium as traveling jam fronts: the wave shears the medium ahead of it into the jammed state, then the medium relaxes behind it. The properties above describe the JAMMED state the wave creates as it passes.

GW lossless conduction (qualitative, no density needed):

```
Whatever the medium density, Z_torsion << Z_matter at any boundary.
Reflectance R ~ (Z_torsion/Z_NS)^2 is negligible for any physically
reasonable density. LIGO's 10+ years of clean GW observations with no
spurious reflections corroborate Z_torsion/Z_matter << 1.
```

3.2 Jobson cell geometry (alpha, phi, r_p -- all CODATA measurements)

The medium's discretization scale is set by the Jobson cell closure condition.
All inputs are CODATA direct measurements -- no cosmological model needed:

```
alpha = 7.2973525693e-3  (QED precision measurements)
phi   = (1+sqrt(5))/2    (geometry)
r_p   = 0.8414 fm        (electron-proton scattering, hydrogen spectroscopy)

N_lock = 2*pi/(alpha*phi) = 532.1 cells per tube circumference
L_J    = alpha*phi*r_p    = 9.93e-18 m = 0.00993 fm
```

This is the Jobson cell edge length -- the smallest discrete arc segment of the (1,2) torus knot of the electron (derived in [DOC ALPHA]). It sets the natural discretization scale of the medium.
The cell's icosahedral shape is NOT assumed -- it is derived in Section 3.2a from the lepton spinor structure of the 2I double group plus the Euler formula.
The golden ratio phi = (1+sqrt(5))/2 appears because the (1,2) Hopf winding vector has norm sqrt(5) = 2*phi-1; the icosahedral vertex positions are parametrized by phi, and the three lepton generations have spinor dimensions 2, 4, 6 summing to 12 = the icosahedral vertex count.
Critically, L_J is scale-invariant: it is a fixed physical constant

(9.93e-18 m) that does not run with probe energy. A nuclear probe at 8 MeV and a hadronic probe at 4 GeV both see the same cell size. This is what makes the N_J regime analysis meaningful: the boundary vs bulk distinction depends only on how many cells fit in the probe's interaction radius, not on the probe energy itself.

3.2a DERIVED LATTICE GEOMETRY FROM FIRST PRINCIPLES (2026-08-22)

The icosahedral geometry of the cell is not assumed -- it is derived from the lepton spinor structure and the Euler formula:

```
Step 1: V = 12 = dim(E+) + dim(G32) + dim(I52) = 2 + 4 + 6
  The three lepton spinors of the 2I double group (electron, muon, tau) have
  spinor dimensions 2, 4, 6. Their sum = 12 = number of cell vertices.
  Both count the C5 rotations of the icosahedral group I (12 C5 elements).

Step 2: E = 30 = 5*V/2 = 30
  The C5 symmetry at each vertex means exactly 5 edges per vertex.
  With V=12: E = 5*12/2 = 30 edges.

Step 3: F = 20 (forced by Euler)
  Euler: V - E + F = 2  =>  F = 2 - 12 + 30 = 20 faces. Not assumed.

Confirmation:
  Maxwell rigidity: 3V - E = 6 = dim(T_1g) + dim(T_2g) = 3 + 3
  The cell is marginally rigid (Maxwell critical) BECAUSE T_1g and T_2g
  are both 3-dimensional. Same number, not a coincidence.
```

COMPRESSION SHELLS (face-adjacency BFS, derived -- ih_lattice_phonon.py IP12-IP14):

```
When one cell is displaced, compression propagates through face-adjacent cells.
Each face has exactly 3 face-adjacent neighbors (since each triangular face
has 3 edges). BFS in the face-adjacency graph (= dual dodecahedron) gives:

  Shell 0: 1  (displaced cell)
  Shell 1: 3  <- directly compressed (3 face-adjacent cells per face)
  Shell 2: 6  <- 3 cells x 2 new edges each
  Shell 3: 6  <- convergence zone
  Shell 4: 3  <- mirror of shell 1
  Shell 5: 1  (antipodal cell)

Compression cloud (shells 1-4): 3 + 6 + 6 + 3 = 18 = n_gravity
n_gravity = 18 = cells that feel the displacement before the antipodal cell.
This is a THIRD independent derivation of n=18 (after dynamical matrix [IP6]
and T_1g/T_2g dimension product [IP8]).

Structure: 1 + 18 + 1 = 20 = F  (total face count).

The 3-9-18 CUMULATIVE pattern:
  Leading hemisphere shells 1+2:  3 + 6 = 9  (= dim T_1g x T_2g = Koide denominator)
  Both hemispheres shells 1-4:  9 + 9 = 18  (= n_gravity)
  Total with center+antipodal: 1 + 18 + 1 = 20  (= F)

[analysis/gravity/ih_lattice_phonon.py, IP12-IP14, 19/19 PASS]
```

3.3 Absolute elastic values: rho_medium = mu_0 (DERIVED, 2026-09-01 update)

[This section originally (2026-08-19) treated the medium density as an open problem and used a cosmological placeholder. That placeholder is superseded below; the original numbers are kept afterward, clearly relabeled, for reference only.]

The medium density is not a free parameter: it follows directly from two results already established elsewhere in this framework -- the EM bulk modulus K = 1/eps_0 (Coulomb's law pressure Green's function; [DOC MAGNETISM] Section 1.2, [DOC HIGGS] Claim C7) and the pressure wave speed v_p = c (Section 3.1, GW170817). The acoustic relation rho = K/v_p^2 gives:

```
rho_medium = 1/(eps_0*c^2) = mu_0 = 1.2566e-6 kg/m^3
[EXACT, CODATA electric constants only, no cosmological model,
 zero free parameters -- P.6b; [DOC MAGNETISM] Section 2.1]
```

With rho_medium = mu_0:

```
Bulk modulus (jammed/wave state): K = rho*(c^2-4/3*Rs^2*c^2) = 1.082e11 Pa
Shear modulus:          G  = rho * Rs^2 * c^2 = 3.576e9 Pa
Young's modulus:        E  = 2G(1+nu) = 1.061e10 Pa
Longitudinal impedance: Z_p = rho * c     = 3.767e2 Pa*s/m
Shear impedance:        Z_s = rho * Rs*c  = 6.704e1 Pa*s/m
[torsion_doc.py T3.5-T3.9; analysis/medium/fluid_vs_jammed_K_resolution.py FJ1-FJ6]
```

NOTE ON K -- TWO REGIMES, NOT ONE VALUE (resolved 2026-09-04): this medium is a non-Newtonian fluid (Section 3.1 above): it flows freely (G -> 0) below the Maxwell-critical shear threshold, and JAMS into a near-incompressible elastic solid only where a wave actively shears it past that threshold. K = 1/eps_0 ([DOC MAGNETISM] Section 1.2) is the STATIC value: a non-propagating point charge never shears the medium, so it sees the unjammed FLUID ground state. The K above is the JAMMED value, relevant to the propagating pressure wave that this section's G, E, Z_p, Z_s already describe -- solved from the general solid relation v_p^2=(K+4/3*G)/rho using rho=mu_0, v_p=c, v_s=Rs*c. The two values differ by exactly 1-4/3*Rs^2 = 4.22%. Verified numerically: each K reproduces v_p=c correctly ONLY in its own regime's formula, and clearly fails (by that same factor) if cross-applied to the other regime's formula -- consistent with these being two distinct mechanical properties of two distinct states of the same medium, not competing measurements of one quantity where one must be wrong. [DOC MAGNETISM]'s K=1/eps_0 is UNCHANGED and correctly scoped to the static/Coulomb regime it actually describes; only this section's own headline (used for wave-propagation quantities) switches to the jammed value.

This reframing surfaces a further, separate question worth its own scrutiny: whether the rho=mu_0 relation above, which itself combines a static-regime K with the propagating wave speed v_p in one formula, mixes the two regimes the same way the old K discrepancy did. Not resolved here -- rho=mu_0 is the framework's single most load-bearing result, cited throughout this and every companion paper, and deserves dedicated scrutiny before any change is made, not a conclusion reached as a side effect of this section's K fix.

That scrutiny surfaces a further fork: the K_jammed value above assumes rho=mu_0 is the quantity held fixed between the static and jammed regimes.

Holding K=1/eps_0 fixed instead and letting rho float gives rho=1.312e-6 kg/m^3 (1.04408*mu_0, +4.408%) -- an equally self-consistent alternative. K/G is EXACTLY 30.25 under both readings (it reduces to (1-4/3*Rs^2)/Rs^2, a function of Rs alone, independent of which absolute quantity is held fixed) -- so no ratio-only check can distinguish them.

A third, micropolar/Cosserat reading (the jammed medium's true shear modulus is zero; today's G is instead a rotational coupling constant alpha) is a genuinely DIFFERENT formula, not just a third value of the same ratio: with no ordinary shear modulus, the classical 4G/3 correction drops out of the longitudinal wave speed entirely, so K stays at the static value K=1/eps_0 with no jammed reduction, and its own analogous ratio is K/alpha = 1/Rs^2 = 31.58 (not 30.25). This offers a real, but modest, piece of evidence against Cosserat: the Weinberg-angle candidate 7/(1+K/G) gives 0.42% error using 30.25 above vs 3.69% using 31.58. That comparison alone tests an isolated component, though -- this framework's own best-precision Weinberg-angle result blends this K/G-dependent "pressure" term with a separate, K/G-independent "vertex" term derived from (1,2) Hopf topology (weighted 1/3 pressure + 2/3 vertex); repeating the comparison on that full blend still favors 30.25 over 31.58 (0.012% vs 1.36% error), but the margin narrows sharply once the K/G-independent 2/3 is included, and the pressure term itself is one candidate among roughly 20 tried in an exploratory scan, not independently established as the correct mechanism. Treat this as real, direction-consistent, but not decisive evidence. Whether the classical solid-elasticity formula used throughout this section applies at all to a granular, contact-jammed medium, rather than requiring a different constitutive law such as this one, remains unresolved either way. The value above is kept as the best-evidenced working default, not a settled derivation.

LEGACY, CONTAMINATED -- NOT PART OF THE DERIVED RESULT (reference only): The version of this section as originally published (2026-08-19) used the working hypothesis rho = rho_Lambda = 5.84e-27 kg/m^3, the Planck 2018 cosmological vacuum density derived from CMB fitting (rho_Lambda = 3*H0^2*Omega_Lambda/(8*pi*G)), before the rho = mu_0 result above existed:

```
G = 1.663e-11 Pa   K = 5.029e-10 Pa
Z_p = 1.752e-18 Pa*s/m   Z_s = 3.117e-19 Pa*s/m

Jeans stability: L_Jeans = Rs*c*sqrt(pi/(G_Newton*rho_Lambda)) = 4725 Mpc
Hubble radius:   c/H0 = 4283 Mpc;  L_Jeans/r_Hubble = 1.10
[torsion_doc.py T3.10-T3.11, legacy reproduction of the as-published numbers]
```

This is CIRCULAR, not merely conditional: rho_Lambda is ITSELF defined via H0 and Omega_Lambda through the Friedmann equation above, so L_Jeans ~ c/H0

follows automatically from that definition -- it is a consistency of the Friedmann equation in a dark-energy-dominated universe, not an independent prediction of the torsion medium (rs_v3.py's own commentary on this same calculation says so explicitly). It offers no evidence for or against the torsion medium and should not be read as corroborating rho_medium = mu_0 above. It is retained only because it was part of the paper as originally published, consistent with how the same rho_Lambda input is labeled elsewhere in the project (medium_properties_clean.py, redshift_doc.py: both "conditional"/"CONTAMINATED", never treated as a derived result).

The medium density is no longer an open problem (resolved by the rho = mu_0 result above); the Jeans-length/Hubble-radius comparison specifically remains unresolved as an open, independent question, since the rho_Lambda input it depended on is circular.

4. JOBSON CELL DISCRETIZATION AND SCALE CORRECTIONS

The four-scale Rs cluster is not random noise. With current PDG 2022 inputs, three of the four scales (nuclear, hadronic, flyby) already agree with Rs to better than 0.1%; the remaining spread is dominated by the galactic scale's known H0-dependence (Section 2.4), not by an unexplained cell-boundary effect:

```
Scale       N_J = R_load/L_J   Regime         Deviation (current PDG 2022)
Nuclear     441                     bulk (N>>1)    -0.09%  (exact Rs)
Galactic    7.86e7                  deep bulk      +2.98%  (H0 systematic)
Flyby       --                      macroscopic    -0.0001%  (geometry)
Hadronic    4.75                    boundary       +0.001%  (exact Rs)
```

[analysis/demos/torsion_doc.py Section 2]

LEGACY NOTE -- why this was believed, and why it is now known wrong (not merely superseded): earlier versions of this document, using an older, less precise pre-PDG-2022 kappa = 0.900 GeV/fm (vs the current PDG 2022 value 0.884 GeV/fm used above), found a +1.81% hadronic deviation and proposed a "Channel A" cell-boundary correction:

```
Rs_eff_had = Rs * (1 + k_A / N_J_had),   N_J_had = R_load_had / L_J
k_A = 12*alpha*(1-alpha*phi^2) = 0.085895
[analysis/rs_scale/rs_v3.py; earlier drafts showed R_load/L_J inverted
 in this formula -- corrected here to match the actual computation]
```

This looked well-supported for two reasons, neither of which holds up: (1) k_A was fit to exactly cancel the old +1.81% gap (k_A = deviation / (L_J/R_load_had), by construction) -- a fit, not evidence of a real boundary effect by itself. (2) A closed form was then found matching that fitted value to -0.12%: 12 icosahedral vertices times the same alpha*phi^2 Fibonacci correction that appears independently in the R9 vev result and the alpha vertex stiffness derivation [analysis/medium/demo/torsion_k_A_derivation.py]. That cross-framework pattern match made

the fit look like more than coincidence.

It is now known to be wrong, not merely unneeded: N_J_had depends only on m_b, alpha, phi, and r_p -- never on kappa. Channel A's own formula therefore predicts a fixed +1.81% hadronic excess that should appear at ANY kappa precision. But the current, more precise PDG 2022 kappa alone (no correction applied) drops the raw deviation to +0.001% (Section 2.3) -- the entire old +1.81% gap is accounted for by kappa's own 1.81% precision change (0.900 -> 0.884 GeV/fm), leaving nothing for a real boundary effect to explain. Applying Channel A's kappa-independent formula to today's best inputs still predicts +1.81%, which the +0.001% measurement directly contradicts.

k_A's closed form is retained here only as a numerically interesting coincidence -- a formula found to match an already-fitted number, built from a real, recurring Fibonacci pattern seen elsewhere in the framework -- not as a validated physical mechanism. It is no longer invoked to explain, predict, or bound any present-day hadronic deviation.

5. MOND PREDICTION AND SPARC COMPARISON

5.1 Derivation of a0

The MOND critical acceleration emerges from the torsion medium as:

```
a0 = v_s * H0 = Rs * c * H0
```

With Rs = sqrt(5)/(4*pi) and Planck 2018 H0 = 67.4 km/s/Mpc:

```
a0 = 0.177941 * 2.998e8 m/s * 2.184e-18 s^{-1}
   = 1.165e-10 m/s^2
```

No free parameters. The MOND acceleration is the product of the torsion wave speed and the Hubble expansion rate -- a resonance between the medium's shear mode and cosmic expansion.

5.1a MOND interpolation function from medium constitutive law [N-3]

The torsion medium has two coupling channels:

```
Bulk (pressure, stiffness K): Newtonian gravity, wave speed c
Shear (stiffness G = K/30.25): low-acceleration coupling, wave speed Rs*c
```

The interpolation function is obtained by combining the two channels in quadrature -- an ansatz chosen for its correct asymptotic limits (Newtonian at high a, shear-dominated at low a), not a first-principles derivation from a full elastic tensor calculation:

```
mu(x) = x / sqrt(x^2 + G/K)  where x = a/a0, G/K = 1/30.25
```

Limits:

```
x >> 1 (a >> a0): mu -> 1              [Newtonian regime]
```

```
x << 1 (a << a0): mu -> x * sqrt(K/G)   [deep shear regime]
```

Transition point: x_t = 1/sqrt(K/G) = 0.1818, mu(x_t) = 1/sqrt(2) exactly.

NEW PREDICTION: in the deep shear regime (a << a0/5.5 = 2.2e-11 m/s^2):

```
v_flat = (G_N*M*a0)^(1/4) * (K/G)^(-1/8) = simple_MOND * (30.25)^(-1/8)
Velocity is 34.7% LOWER than simple MOND (mu(x) = x/sqrt(1+x^2)) predicts.
Testable in ultra-diffuse galaxies (UDGs) and stellar streams at very low a.
```

The torsionverse mu(x) shape follows from K/G = 30.25 (T3.2) once the quadrature ansatz above is granted; K/G itself has zero free parameters, but the quadrature combination rule is an assumed functional form, not a separately derived one.
Script: analysis/gravity/mond_interpolation.py (8/8 PASS, T5.3-T5.4)

5.2 Comparison with McGaugh et al. 2016

McGaugh, Lelli, and Schombert (2016) measured the Radial Acceleration Relation (RAR) for 153 SPARC galaxies. Their best-fit a0 = 1.20e-10 m/s^2, with a quoted 1-sigma uncertainty of 0.02e-10 m/s^2 (1.7%).

Using an intermediate H0 = 70.0 km/s/Mpc (between the Planck 67.4 and local 73.3 values used elsewhere in this document, and a common compromise value in the literature):

```
Predicted:  a0 = Rs*c*H0 = 1.2102e-10 m/s^2
Measured:   a0 = 1.2000e-10 m/s^2
Difference: +0.85%  (+0.51 sigma at the RAR's own quoted 1.7% uncertainty)
Free parameters adjusted: zero
```

[analysis/gravity/mond_rotation.py, Part A]

At the Planck H0 = 67.4 value used in Section 5.1, the same comparison gives a0 = 1.165e-10 m/s^2, a larger -2.9% difference (1.7 sigma at the same 1.7% uncertainty) -- the H0 choice matters at this level of precision (see Section 2.4's discussion of the same H0-dependence at the galactic scale). The 0.51 sigma headline figure specifically uses H0 = 70.0, not Planck's 67.4.

No interpolation function is assumed; a0 is the single derived parameter.

5.3 Redshift evolution prediction

If the medium is cosmologically coupled (v_s = Rs*c at all epochs), then a0 evolves with the Hubble rate:

```
a0(z) = Rs * c * H(z)
```

At z=1 (using Planck 2018 LambdaCDM: H(z=1)/H0 = sqrt(0.315*8 + 0.685) = 1.790):

```
a0(z=1) = 1.790 * a0(z=0) = 2.086e-10 m/s^2
```

This is a falsifiable prediction testable with JWST extended rotation curves at z > 0.5. A constant a0 across redshift would require framework modification.

6. FALSIFIABLE PREDICTIONS

6.1 MOND acceleration at z=1

```
a0(z=1) = 2.086e-10 m/s^2   (vs a0(z=0) = 1.165e-10 m/s^2)

Testable: JWST extended rotation curves at z ~ 0.5-2.
Confirm: galaxies at z~1 show flat curves transitioning at 2.09e-10 m/s^2.
Falsify: a0 is constant across redshift to 5% precision.
```

6.2 B meson medium radius

The torsion medium framework predicts the effective interaction radius of the B meson through the medium's response to the b-quark mass:

```
r_B = Rs * m_b / kappa
    = 0.177941 * 4180 MeV / (0.884 GeV/fm)
    = 0.826 fm
```

This is comparable to the proton charge radius (0.841 fm) -- a non-trivial prediction since the B meson's quark content is very different from the proton. Not yet measured; testable at the Electron-Ion Collider or by lattice QCD B-meson form factor calculations. Current framework prediction:

```
kappa_pred = Rs * m_b / r_p = 0.884 GeV/fm   (matches PDG to 0.0%)
```

Note: kappa_pred above is the same kappa*r_p/m_b = Rs identity as Section 2.3, algebraically solved for kappa instead of Rs -- it is a restatement, not an independent confirmation. The r_B prediction above it (using the B meson's own, distinct, not-yet-measured radius rather than r_p) is the genuinely new, independently testable claim of this subsection.

6.3 Anisotropy of Rs/alpha

The ratio Rs/alpha = 24.38 equals the Chern-Simons coupling Q = 4*pi^2/phi = 24.40 to within 0.06% [DOC ALPHA]. This means the matter/EM anisotropy of the medium is itself derivable from the (1,2) Hopf geometry. The prediction is that no new physical constant enters the relationship Rs/alpha -- it is determined by the same topological input as alpha itself. This is testable if either Rs or alpha is measured with significantly improved precision.

6.4 Jupiter flyby dV

```
K_Jupiter / K_Earth = 4.28x  (from saturation model)
```

A spacecraft flyby of Jupiter at < 5 R_J with modern Doppler tracking would give dV ~ 50-200 mm/s, well above thermal noise. This is the cleanest geometric test of the K-formula because Jupiter's large size and rotation rate maximize the cos(dec) asymmetry. Confirm: dV matches K_Jupiter prediction. Falsify: dV < 5 mm/s for close Jupiter flyby at < 5 R_J.

7. CROSS-SCALE CORROBORATION

7.1 Constitutive law consistency at three scales

At each of three independent scales (particle, hadronic, galactic), the medium obeys a zero-then-constant constitutive law: below a threshold the medium response is zero; above it the medium saturates to a constant Rs-determined ratio. This pattern is the signature of a jammed medium (Liu and Nagel 1998):

```
Particle scale (g-factor): electron g-2 shows no medium correction below
L_J = 0.00993 fm -- the QED vacuum behaves as free space at sub-cell scales.
Above L_J the icosahedral vertex scattering contributes (the alpha derivation
in [DOC ALPHA]).

Hadronic scale (string tension): below Lambda_QCD the string tension is zero;
above it, kappa = constant * Rs (the hadronic saturation floor).

Galactic scale (rotation curves): below a0 = Rs*c*H0 Newtonian gravity
applies; above the threshold it is modified (the MOND regime).
```

The same constitutive structure at three independent scales spanning 30 orders of magnitude is strong qualitative evidence for a single underlying medium.

7.2 PSR B1828-11 pulsar corroboration

The pulsar PSR B1828-11 shows periodic modulations in its spin-down rate and beam geometry. The ratio of its two principal modulation periods:

```
P1/P2 = 511 days / 256 days = 1.996
```

This is consistent with the (1,2) torus knot winding ratio n=2 to 0.07 sigma (integer n=2) and 0.41 sigma (n_exact = 2.01869). The physical interpretation is that the pulsar's beam geometry traces a (1,2) winding -- the same topology as the electron in the torsion medium framework.

This is cross-scale corroboration, not independent evidence: the measurement precision (unc ~ 0.055) is insufficient to distinguish n=2 from n_exact.
A 3x improvement in timing would resolve this. Both n=2 and n_exact are consistent with current data.
[analysis/alpha/pulsar_double_spin.py]

7.3 The Jobson-Higgs conjecture

Higgs (1964) correctly identified that a scalar field permeating the vacuum gives particles mass through spontaneous symmetry breaking. The Standard Model treats the Higgs boson mass m_H as a free parameter -- it is measured but not predicted. This framework identifies the geometric origin of that mass scale.

The Jobson cell energy is:

```
E_cell = 2*pi*hbar*c / L_J = N_lock * hbar*c / r_p = 124.799 GeV
```

The Higgs boson is spin-0 (scalar). A scalar particle couples to the EM field via charge^2 (two EM vertices), giving leading QED radiative corrections of order alpha/pi -- exactly twice the Schwinger fermion correction alpha/(2*pi). This is standard QED for scalars, not a fit.

The Jobson-Higgs conjecture:

```
m_H = E_cell * (1 + alpha/pi) = 125.089 GeV
```

Comparison with measurement:

```
vs PDG combined (older, 125.09 GeV): -1.18 MeV = -0.001%
vs PDG 2022 (125.20 +/- 0.11 GeV):  -111 MeV  = -1.01 sigma
```

Framing: Higgs found the mechanism. This work finds a geometric energy scale that coincides with the Higgs mass. The Jobson cell is a pure geometric structure -- the icosahedral unit cell of the torsion medium -- whose characteristic energy E_cell = 124.8 GeV happens to match m_H to within the scalar QED radiative correction alpha/pi. This conjecture has since been substantially strengthened [DOC HIGGS]: the SSB mechanism is derived (jamming = spontaneous symmetry breaking); the gauge coupling T_1g = W/Z is derived; the Weinberg angle's one-loop result is well-supported (-0.91 sigma; a previously-claimed two-loop term is RETRACTED -- see [doc_higgs] Claim 11 for the full status). The remaining open question is the field identity -- whether L_HWW = alpha^2*phi^2*|H|^2*|W|^2 (its FORM derived from the cell, its coefficient NOT yet independently derived -- see [doc_higgs] Section 5a.3) IS the SM Higgs Lagrangian, or whether the cell is the geometric origin of the SM Higgs field without being identical to it. The mass, coupling structure, and SSB mechanism are derived; the field identification is interpretive, not axiomatic.

The core direction-independent relation:

```
m_H * r_p = 2*pi*hbar*c*(1+alpha/pi) / (alpha*phi)
```

connects three independently measured constants -- m_H (electroweak), r_p (QCD), alpha (EM/topology) -- implying that m_H and r_p are not independent.

Sub-cell results [DOC HIGGS]: for particles smaller than the cell (N_J < 1: Higgs, W, Z, top), the relevant coupling is the bulk medium Poisson ratio nu, not vertex geometry. The Higgs quartic self-coupling is:

```
lambda = (1-nu)/4 = 2*pi^2/(16*pi^2-5) = 0.12909
```

Derived from Rs alone via 1-nu = 1/(2*(1-Rs^2)). This gives:

```
- lambda_SM (PDG 2022) = 0.12928: gap = 0.149% = 0.84 sigma. CONSISTENT.
- vev: v = m_H / sqrt(2*lambda) = 246.185 GeV vs v_EW = 246.22 GeV  (-35 MeV = -0.014%).
    With a candidate alpha^2*phi^2 term (coefficient NOT independently
    derived, see [doc_higgs] Section 5a.3): v* = 246.219 GeV (EW gap: -0.7 MeV).
- Gamma_H = alpha * Rs * m_H / (4*pi^2) = 4.114 MeV (+1.1% from PDG 4.07).
```

The Gamma_H formula follows from the alpha self-consistency equation:

```
Q*alpha = Rs (leading order)  =>  alpha/phi = Rs/CS
```

```
=>  Gamma_H = alpha*(alpha/phi)*m_H = alpha*Rs*m_H/(4*pi^2)
```

Fermion masses (essentially derived): m_e closed (0.000069%); m_mu/m_tau essentially derived via R=(phi^6-1)*(1-alpha)+Koide (0.18% residual). Full Yukawa hierarchy (heavier generations) deferred to fermion mass paper.

A previously-claimed closure of the cell binding energy via jamming stiffness [DOC HIGGS, Claim 8] -- 7*k_n_max/(2*pi)=1+alpha+alpha^2*phi -- is RETRACTED (2026-09-29); both k_n_max and the alpha^2*phi term were found unsupported, and a broader closed-book review found general mathematical obstructions to any such route. E_cell's own value is unaffected -- see [DOC HIGGS] Section 8 and [DOC ALPHA] for full detail.

The Clebsch-Gordan decomposition T_1g x T_1g = A_g + T_1g + H_g in I_h (A_g appears exactly once by Schur's lemma) gives a unique Higgs-gauge coupling FORM [DOC HIGGS]; the candidate coefficient L_HWW = alpha^2*phi^2*|H|^2*|W|^2 is NOT independently derived (see [doc_higgs] Section 5a.3) and extends m_H = E_cell*(1+alpha/pi+alpha^2*phi^2) = 125.106 GeV as an unverified candidate, numerically closing the vev gap to -0.7 MeV. H->WW (T_1g x T_1g -> A_g, CG=1) and H->bb (G_g x G_g -> A_g, CG=1) selection rules are structurally derived (multiplicity/form only). Forbidden prediction: H -> T_1g + T_2g (T_1g x T_2g has no A_g component).

The vertex stiffness ratio is derived from the Born amplitude + one-loop EM self-energy balance [DOC ALPHA, Section 4.5]:

```
k_n*(1+alpha) = alpha*phi*k_LW
=> k_n/k_eff = alpha*phi/(1+alpha*phi^2)  [Fibonacci phi^2=phi+1, exact]
=> alpha (from quadratic) = 7.2973525856e-3  vs CODATA 7.2973525693e-3
Residual 0.00000022%  -- essentially closes the last open step in [DOC ALPHA].
```

Same Fibonacci factor (1+alpha*phi^2) governs BOTH the alpha vertex renormalization and the Higgs vev correction alpha^2*phi^2.

Full cell geometry reference: [DOC JOBSON CELL], https://doi.org/10.5281/zenodo.22032906.

Script: torsion_doc.py (Sec 7.3: Jobson-Higgs conjecture); detailed Higgs results in analysis/demos/higgs_doc.py.

7.4 L_J as QCD UV cutoff: asymptotic freedom from cell geometry

The Jobson cell length L_J = 0.00993 fm is the UV cutoff of the torsion medium lattice. The QCD confinement scale Lambda_QCD^{-1} ~ 1 fm is the IR scale -- these are naturally 99x apart, not a gap to close.

The physical picture:

```
At E < E_cell (~125 GeV): the medium scatters -- QCD confinement is active.
At E > E_cell (~125 GeV): above the cell Nyquist frequency -- the medium
becomes transparent. The coupling decreases: asymptotic freedom.
```

This means E_cell = 2*pi*hbar*c/L_J is BOTH the Higgs boson mass (Jobson-Higgs

conjecture, Section 7.3) AND the QCD asymptotic freedom onset. The electroweak scale and the QCD UV cutoff are the same object: the Jobson cell energy.

Lambda_QCD sits at N_J ~ 99 cells -- the deep bulk regime of the cell lattice (T2.5), where the medium scatters uniformly and confinement is the result. An older, pre-PDG-2022 hadronic Rs deviation of +1.81% was once attributed to hadronic probes (N_J ~ 4.75) being near the cell boundary; with current PDG 2022 values the hadronic deviation is +0.001% (Section 4), so this specific number no longer illustrates a boundary effect -- the regime classification itself (bulk vs. near-boundary N_J) is unaffected.

The cell geometry therefore explains three QCD phenomena from one object:

```
1. Confinement (below E_cell, medium scatters)
2. Asymptotic freedom (above E_cell, medium transparent)
3. Hadronic Rs deviation (near-cell boundary effects, T2.5)
```

QCD/EW beta function implications [DOC HIGGS]:

```
The N_J regime transition provides the physical mechanism for asymptotic freedom:
  Bulk regime (N_J > 1, E < E_cell/(2*pi) ~ 20 GeV): vertex stiffness
    coupling -> strong at low E, confinement tendency [correct for QCD]
  Sub-cell regime (N_J < 1): Poisson coupling (soft) -> weak at high E
    [asymptotic freedom direction] [correct for QCD]
  W/Z bosons: permanently sub-cell (N_J ~ 0.25) -> never enter vertex
    regime -> no confinement [correct: EW is not confining]
The beta function SIGNS for all three SM forces follow from the N_J
regime structure without invoking detailed loop calculations.
The Weinberg angle connection: from the medium's mechanical properties,
  7 * G/(K+G) = 7 * (1/(1+K/G)) = 7/(1+30.25) = 0.224
  vs sin^2(theta_W) = 0.223  (0.4% off)
The factor 7 = dim(A_g)+dim(T_1g)+dim(T_2g) = 1+3+3 (EM-coupled sector).
This connects K/G = 30.25 directly to the Weinberg angle.
```

8. HONEST STATUS ASSESSMENT

[EMPIRICAL, zero free parameters, no cosmological model]:

```
- Rs at nuclear scale: 8/sigma_piN = 0.17778, deviation -0.09%
- Rs at flyby scale: c_torsion = Rs*c to 0.0001%
- Rs at hadronic scale: kappa*r_p/m_b = 0.17794, deviation +0.001% (PDG 2022)
- a0 = Rs*c*H0 = 1.165e-10 m/s^2 (Planck H0); 0.51 sigma from 153 SPARC
  galaxies using H0=70.0 (Section 5.2, analysis/gravity/mond_rotation.py)
- nu = 0.4837, K/G = 30.25, Z_s/Z_p = Rs  (from v_p, v_s only)
- L_J = 9.93e-18 m, N_lock = 532.1  (from CODATA alpha, r_p)
- Medium is a non-Newtonian (dilatant) liquid: jams under shear
```

[LEGACY, contradicted by current PDG 2022 values, not merely superseded (see Section 4 for the full reasoning)]:

```
- Rs at hadronic scale (pre-PDG-2022): +1.81% deviation; a "Channel A"
  correction k_A = 12*alpha*(1-alpha*phi^2) = 0.08590 (-0.12%) was proposed.
  Its N_J_had term depends only on m_b/alpha/phi/r_p, never on kappa, so
  it predicts a fixed +1.81% excess at any kappa precision -- contradicted
  by the current PDG 2022 kappa, which gives +0.001% directly, no
  correction needed or supported.
```

[EMPIRICAL, with H0 systematic]:

```
- Rs at galactic scale: +2.98% deviation (Planck H0) / -5.31% (local H0),
  consistent with the ~8% CMB-vs-local H0 tension (Section 2.4)
```

[DERIVED (2026-08-22 update; corrected 2026-09-02, see Section 3.1)]:

```
- v_p = c: A_g isotropic compression mode has C5 char = +1 (trivial, totally
  symmetric -> no dispersion -> propagates at c). GW170817 confirms.
```

```
  [Section 3.1] T_1g (photon/W/Z) is a separate transverse vector channel,
  also undamped at c via its own constructive C5 char=+phi, but not the
  v_p/compression branch itself.
- v_s = Rs*c: T_2g shear mode has C5 char = -1/phi (destructive/shear coupling
  -> dispersive -> propagates at Rs*c). Flyby K-formula confirms. [Section 3.1]
Both wave speeds are now identified with specific cell modes, not assumed.
```

[DERIVED (2026-08-23 update -- cell structure, dihedral, phonon picture)]:

```
Complete icosahedral cell structure verified by script (JC1-JC9, 43/43 PASS):
- 20 triangular faces, inradius = phi^2/sqrt(3) in standard coords [JC2]
- Sum of 20 outward face normals = 0: A_g phonon is radially symmetric [JC3]
- Icosahedral face-to-face dihedral = arccos(-sqrt(5)/3) = 138.19 deg [JC5]
  (NOT 116.57 deg -- that is the DODECAHEDRON dihedral, the dual solid) [JC6b]
- Tau corkscrew Hamiltonian cycle: exists, 20 steps, all at 138.19 deg [JC7-JC9]
- Cell is a spring lattice, not oscillator. A_g zone-boundary phonon = Higgs.
- W/Z = directed T_1g transverse oscillations (3 orthogonal axes, vector
  mode); A_g = isotropic compression cone (the v_p/GW170817 branch).
- Wave linger = entrainment force: see PHYSICAL MODEL section above.
- Flex/jamming unified: geometric (3V-E=6) and elastic yield are the same
  event at Maxwell criticality. [DOC JOBSON CELL; jobson_cell_doc.py 47/47]
```

[DERIVED (2026-08-22 update -- crystal field projection)]:

```
When Jobson cells (I_h medium) are embedded in a macroscopic crystal with lower
symmetry, the external crystal field lifts I_h degeneracies. The higher-dimensional
I_h irreps split under the crystal symmetry group. Key result for superconductivity:
H_g (dim=5, l=2 d-orbitals) under D4h (square-planar CuO2 crystal field) decomposes
as A1g + B1g + B2g + Eg. The B1g component = d_{x^2-y^2} = d-wave gap symmetry.
In I_h-symmetric systems (K3C60): A_g channel wins -> s-wave superconductor.
In D4h crystal field (YBCO, BSCCO): H_g/B1g channel selected -> d-wave.
Same T_1u x T_1u CG product; crystal field selects which I_h component dominates.
[crystal_field_projection.py 9/9 PASS; doc_entanglement Prediction (b)]
```

[PREDICTIONS, not yet tested]:

```
- a0(z=1) = 2.086e-10 m/s^2 (JWST rotation curves)
- r_B = 0.826 fm (EIC / lattice QCD B-meson form factor)
- Jupiter flyby dV ~ 50-200 mm/s at < 5 R_J
```

What this paper does NOT claim:

```
- That the torsion medium is a proven physical entity (it is an inference
  from four empirical coincidences plus GW170817 and Planck 2018)
- That the MOND effect is fully explained (only a0 is derived; the
  interpolation function mu(x) = x/sqrt(x^2+G/K) is NOW derived
  from K/G = 30.25 -- mond_interpolation.py 8/8 PASS, N-3; the quadrature
  combination rule itself is an ansatz, not separately derived)
- That k_A = 12*alpha*(1-alpha*phi^2) is axiomatically proven. The formula
  matches to -0.12%; the physical argument (12 icosahedral vertices × alpha ×
  Fibonacci correction) is compelling but not yet a formal topological proof.
- That the Jobson-Higgs mechanism constitutes an axiomatic mathematical proof.
  All core claims (m_H, v, lambda, Weinberg angle, SSB=jamming, Gamma_H) are
  derived with zero free parameters and documented in [DOC HIGGS]. The derivation
  is physics-level, not a formal axiom proof. The field identity -- whether the
  cell's Lagrangian L_HWW IS the SM Higgs Lagrangian -- remains interpretive.
```

9. CONCLUSION

The ratio Rs = sqrt(5)/(4*pi) = 0.17794 appears at four independent experimental scales with a cluster mean deviation of +0.72% and 1.30% spread, with zero free parameters adjusted. The four appearances are: nuclear binding (-0.09%), hadronic string tension (+0.001%, current PDG 2022), galactic MOND acceleration (+2.98% at Planck H0, -5.31% at local H0, tracking the known H0 tension), and spacecraft flyby K-formula (0.0001%).

From two direct observational inputs (v_gw = c from GW170817; c_torsion = Rs*c from flyby anomaly) model-independent properties of the torsion medium are

derived: nu = 0.4837, K/G = 30.2, Z_s/Z_p = Rs. The MOND critical acceleration a0 = Rs*c*H0 = 1.165e-10 m/s^2 agrees with the 153-galaxy SPARC RAR to 0.51 sigma.

Three falsifiable predictions distinguish this framework from alternatives:
(1) a0 increases with redshift as H(z), reaching 2.086e-10 m/s^2 at z=1
[under cosmological H(z); see note below];
(2) the B meson interaction radius is r_B = 0.826 fm;
(3) a close Jupiter flyby should show dV ~ 50-200 mm/s.

Note on prediction (1): This assumes standard cosmological coupling -- H0 is the cosmic expansion rate and a0 = Rs*c*H(z) evolves with it. An alternative interpretation being investigated ([DOC ORBIT PRESSURE]) treats redshift as locally generated by stellar/planetary boson emission, in which case H would be approximately constant across z and a0 would be constant. The JWST measurement therefore distinguishes two interpretations simultaneously: a0 increasing with z confirms cosmological coupling; a0 constant confirms the local-boson picture. Either outcome is informative for this framework.

The geometric origin of Rs from the (1,2) Hopf fibration is established in the companion paper [DOC ALPHA].

COMPANION SCRIPTS (reproducibility):

```
analysis/demos/torsion_doc.py        -- MAIN: all key results, all sections  [STANDALONE -- no external deps]
analysis/gravity/ih_lattice_phonon.py -- derived lattice: V/E/F, shells, n=18; 19/19 PASS
analysis/gravity/mond_rotation.py    -- 0.51 sigma RAR comparison (Section 5.2)
analysis/gravity/mond_interpolation.py -- MOND interpolation function (Section 5.1a)
analysis/alpha/pulsar_double_spin.py -- PSR B1828-11 corroboration (Section 7.2)
analysis/quantum/face_gluon_geometry.py -- FG5, T_2g shear amplitude (Section 3.1)
Repository: https://doi.org/10.5281/zenodo.22016573
```

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Repository: https://doi.org/10.5281/zenodo.22016573

Code: https://github.com/Destraag/torsionverse_public